Phase 1 : Object Storage Foundation

1A

```
divyanshB@divyanshB-ubuntu:~/PES1UG24AM094-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe
9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
divyanshB@divyanshB-ubuntu:~/PES1UG24AM094-pes-vcs$
```

1B

```
All Phase 1 tests passed.
divyanshB@divyanshB-ubuntu:~/PES1UG24AM094-pes-vcs$ find .pes/objects -type f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
divyanshB@divyanshB-ubuntu:~/PES1UG24AM094-pes-vcs$
```

Phase 2 : Tree Objects

2A

```
divyanshB@divyanshB-ubuntu:~/PES1UG24AM094-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
divyanshB@divyanshB-ubuntu:~/PES1UG24AM094-pes-vcs$
```

2B

```
divyanshB@divyanshB-ubuntu:~/PES1UG24AM094-pes-vcs$ xxd $(find .pes/objects -type f | head -1) | head -5
00000000: 626c 6f62 2031 3500 5465 7374 2069 6e74  blob 15.Test int
00000010: 6567 7269 7479 0a                        egriXy.
divyanshB@divyanshB-ubuntu:~/PES1UG24AM094-pes-vcs$
```

Phase 3 : The Index (Staging Area)

```
Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:   object.c
  untracked:   test_tree.c
  untracked:   tree.c
  untracked:   index.c
  untracked:   pes.c
  untracked:   Makefile
  untracked:   test_objects
  untracked:   test_sequence.sh
  untracked:   .gitignore
  untracked:   commit.h
  untracked:   index.h
  untracked:   README.md
  untracked:   commit.c
  untracked:   test_objects.c
  untracked:   test_tree
  untracked:   tree.h
  untracked:   pes.h
```

3A

3B

```
divyanshB@divyanshB-ubuntu:~/PES1UG24AM094-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4  1776754737 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f  1776754737 6 file2.txt
divyanshB@divyanshB-ubuntu:~/PES1UG24AM094-pes-vcs$
```

Phase 4 : Commits and History

4A

```
divyanshB@divyanshB-ubuntu:~/PES1UG24AM094-pes-vcs$ ./pes commit -m "Initial commit"

echo "world" >> file1.txt
./pes add file1.txt
./pes commit -m "Add world"

echo "farewell" >> file2.txt
./pes add file2.txt
./pes commit -m "Add farewell"

./pes log
Committed: ddfac0e6fb25... Initial commit
Committed: 199f47dc59a0... Add world
Committed: 799d36088da9... Add farewell
commit 799d36088da98098335da60fe55094d30263d94750ca313ff6eaea4ac90674e5
Author: PES User <pes@localhost>
Date:   1776755022

    Add farewell

commit 199f47dc59a082354e5b91ab7bfb7542c15b612d220307bc14d919d7194c2806
Author: PES User <pes@localhost>
Date:   1776755022

    Add world

commit ddfac0e6fb25c19dfbba9c8213ac2242fa475e20bb5d40a10b64f09ac3dd1f5c
Author: PES User <pes@localhost>
Date:   1776755022

commit ddfac0e6fb25c19dfbba9c8213ac2242fa475e20bb5d40a10b64f09ac3dd1f5c
Author: PES User <pes@localhost>
Date:   1776755022

    Initial commit
```

4B

```
divyanshB@divyanshB-ubuntu:~/PES1UG24AM094-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/01/17c3198da3f6c59b379d2b842eaaf151f104cab664cb090a96877dabc21573
.pes/objects/19/9f47dc59a082354e5b91ab7bfb7542c15b612d220307bc14d919d7194c2806
.pes/objects/75/f65e442b90c28357c47f4a2ef37ae3d1a29d8d8487b4d545f78d10d5e7dd3e
.pes/objects/79/9d36088da98098335da60fe55094d30263d94750ca313ff6eaea4ac90674e5
.pes/objects/94/81d1464f27b48af9aa2fca9a56e7789f744b2a9158e20e6ef96f5d1c59c518
.pes/objects/98/d5ec3c714448caebca62f7f76c0a8345bdb07ed99b1fcbebc58d8f6afefefa
.pes/objects/c0/f44fcbb63d68bf6bdeadb5b1b74630bb94a4e6d8d942930e47bf872938cb30
.pes/objects/ca/b4edff685f6b5cd7197f73179839ca37a821be03a84e0fa10a82d74c12ab14
.pes/objects/dd/fac0e6fb25c19dfbba9c8213ac2242fa475e20bb5d40a10b64f09ac3dd1f5c
.pes/objects/fe/76325aa5521b207ebe01e12fd8e9e3abf030cacd5398e3744a3a56a81ad1bd
.pes/refs/heads/main
divyanshB@divyanshB-ubuntu:~/PES1UG24AM094-pes-vcs$
```

4C

```
divyanshB@divyanshB-ubuntu:~/PES1UG24AM094-pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
799d36088da98098335da60fe55094d30263d94750ca313ff6eaea4ac90674e5
ref: refs/heads/main
```

Final Integration

```
divyanshB@divyanshB-ubuntu:~/PES1UG24AM094-pes-vcs$ ./test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)

--- First Commit ---
Committed: e81a5ffd623a... Initial commit

Log after first commit:
```

```
--- First Commit ---
Committed: e81a5ffd623a... Initial commit

Log after first commit:
commit e81a5ffd623a871ee1e71b8a429c479ad522558ba57d3f6de5a85c83032ef7f5
Author: PES User <pes@localhost>
Date:   1776755319

    Initial commit


--- Second Commit ---
Committed: 54af20c37fa0... Update file.txt

--- Third Commit ---
Committed: 5924d6a357b1... Add farewell

--- Full History ---
commit 5924d6a357b1e268567915338447f253cb5034bef08a3e63ae1293f2a2d87aa2
Author: PES User <pes@localhost>
Date:   1776755319

    Add farewell

commit 54af20c37fa0e9f26bd17359b473305c9eb6c5539ad38635094878354e7c63c3
Author: PES User <pes@localhost>
Date:   1776755319

    Update file.txt

commit e81a5ffd623a871ee1e71b8a429c479ad522558ba57d3f6de5a85c83032ef7f5
Author: PES User <pes@localhost>
```

```
    Update file.txt

commit e81a5ffd623a871ee1e71b8a429c479ad522558ba57d3f6de5a85c83032ef7f5
Author: PES User <pes@localhost>
Date:   1776755319

    Initial commit

--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
5924d6a357b1e268567915338447f253cb5034bef08a3e63ae1293f2a2d87aa2

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/54/af20c37fa0e9f26bd17359b473305c9eb6c5539ad38635094878354e7c63c3
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39220ef68d4c7c7e0d6695797533f50d
.pes/objects/59/24d6a357b1e268567915338447f253cb5034bef08a3e63ae1293f2a2d87aa2
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/e8/1a5ffd623a871ee1e71b8a429c479ad522558ba57d3f6de5a85c83032ef7f5

=== All integration tests completed ===
divyanshB@divyanshB-ubuntu:~/PES1UG24AM094-pes-vcs$
```

Phase 5 & 6: Analysis-Only Questions

Q5.1 Implementation of checkout: To implement pes checkout <branch>, I would update the HEAD file to point to the new branch reference. The working directory must then be updated by reading the Tree object associated with the target commit. This is complex because it requires a safe diff: deleting files not in the target tree, adding missing ones, and updating modified ones without losing uncommitted user data.

Q5.2 Dirty Directory Detection: I would compare the current working directory's mtime and size (via stat) against the entries in the index. If they differ, the directory is "dirty." To check against the branch, I would compare the index hashes against the hashes in the target commit's Tree object.

Q5.3 Detached HEAD: If a commit is made in a detached HEAD state, the new commit's parent is the previous hash, but no branch reference (like main) is updated. To recover these, a user would need to find he orphaned commit hash in the object store and manually point a new branch at it.

Q6.1 Reachability Algorithm: I would use a Mark-and-Sweep algorithm.

Mark: Start from all references in refs/heads/ and HEAD, recursively following every tree and blob hash.

Sweep: Iterate through all files in .pes/objects/ and delete any file whose name (hash) is not in the "reachable" set.

Q6.2 Race Conditions in GC: If GC runs while a commit is happening, GC might see a newly written blob that isn't yet referenced by a commit object and delete it. Git avoids this using a grace period (not deleting recent objects).